ALICE L WALTON SCHOOL OF MEDICINE

Hallmarks of Neoplasia: Genomic Instability

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Adapted from: DNA Mutation and Repair

Listen to this Brick
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Learning Objectives (2) Versions

After completing this brick, you will be able to:

1. Describe the relationship between DNA damage and genomic instability.
2. Recognize human developmental disorders and cancer predisposition syndromes associated with defects in nucleotide excision rep[illegible]bination, and nonhomologous end joining.

CASE CONNECTION

You are on your dermatology rotation, and your first patient of the day exhibits extensive freckling of the face and neck with very few freckles on the shoulders. They remark that that they have had two tumors removed from their face that turned out to be squamous cell carcinoma.

What syndrome do you suspect in this patient? What is the underlying mechanism? Consider your answer as you read, and we'll revisit the questions at the end of the brick.

GO TO CONCLUSION ↓

As we continue our journey through the hallmarks of neoplasia, we arrive at genomic instability (Figure 1).

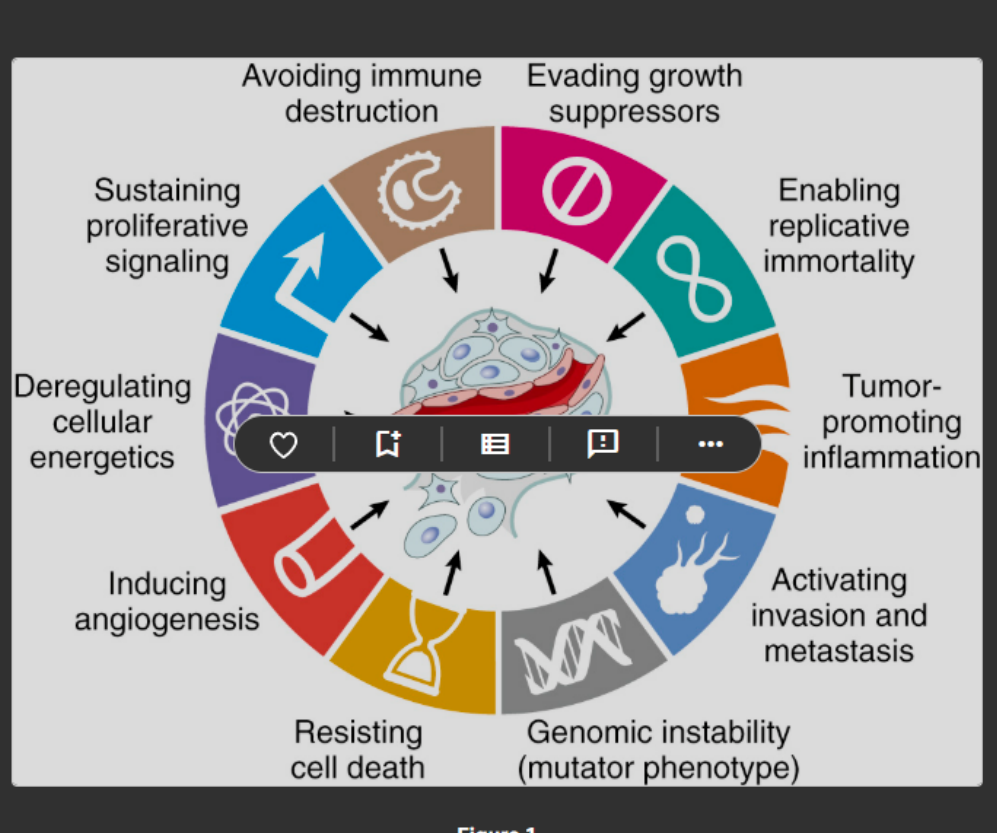

Avoiding immune destruction
Evading growth suppressors
Sustaining proliferative signaling
Enabling replicative immortality
Deregulating cellular energetics
Tumor-promoting inflammation
Inducing angiogenesis
Activating invasion and metastasis
Resisting cell death
Genomic instability (mutator phenotype)

Figure 1

Genomic instability refers to increased propensity for mutations, including DNA base changes and structural alterations, in the genomic DNA. People with deficient DNA repair mechanisms, or those with increased exposure to sources of genetic damage, will have increased genomic instability.

What Are DNA Mutation and Repair (LO #1)?

DNA is much more fragile and prone to chemical damage than you might expect. It's estimated that every cell in the human body (all 30-50 trillion of them) incurs around 70,000 DNA lesions per day. A DNA lesion is a broad term that refers to any site of damage within a polynucleotide, possibly resulting in a strand break, base modification, base deletion, or base insertion.

To handle the burden of constant genetic damage, there are several different biochemical repair mechanisms for the maintenance of genomic stability. The stakes are high. The awe-inspiring complexity through which our bodies function is susceptible to dysfunction at every level. Even a simple alteration to a single gene can hav[illegible]s filled with diseases where small changes in specific proteins manifest as catastrophic or debilitating disorders. As future physicians, we must understand the causes and consequences.

Review: What is the most commonly mutated gene in cancer, also known as the "guardian of the genome"?

How Do Mutations Occur?

When a DNA lesion is not corrected and causes a change in the DNA base sequence, it is referred to as a mutation. A mutation is retained in all subsequent generations of the cell's lineage.

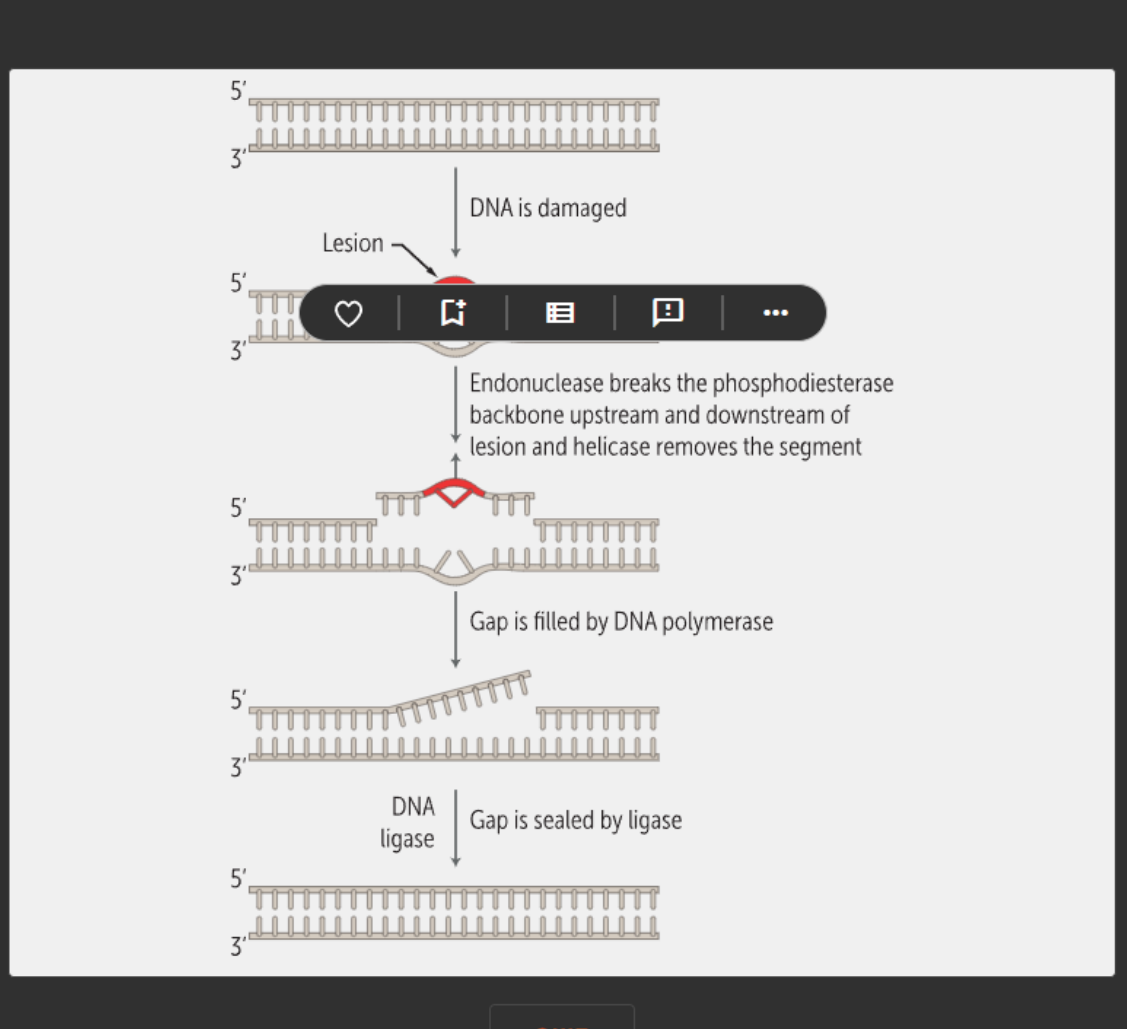
Cellular processes keep a vigilant eye on the integrity of our DNA. Repair mechanisms are constantly at work correcting any DNA lesions. There are even dedicated times when cells will conduct global assessments of their genome. Remember the phase[illegible]heckpoints are established within all replicating cells, especially before and after DNA replication (S phase). Cells will halt their division if a high level of damage is detected and attempt DNA repair. However, if too much damage has been sustained, special mechanisms for apoptosis (programmed cell death) will be activated.

How Is DNA Repaired (LO #2)?

Repair mechanisms tend to address select types of DNA lesions. We will focus our attention below on select DNA repair mechanisms associated with cancer predisposition syndromes, the specific names of which will be discussed in each section.

Xeroderma Pigmentosum: A Defect in Nucleotide Excision Repair

Nucleotide excision repair (NER), a "remove and replace" mechanism, is capable of recognizing a broader range of helix-distorting single-strand lesions. Instead of a single base, a stretch of nucleotides containing the lesion is removed during NER (Figure 2). NER relies on special endonucleases to sever the phosphodiester backbone upstream and downstream of the lesion. DNA polymerase and ligase then fill and seal the created gap.


5'
3'
DNA is damaged
Lesion
Endonuclease breaks the phosphodiesterase backbone upstream and downstream of lesion and helicase removes the segment
Gap is filled by DNA polymerase
DNA ligase
Gap is sealed by ligase

QUIZ

Figure 2

NER is particularly important for correcting UV-light induced damage (pyrimidine dimers). Because humans lack photolyase, NER is an important mechanism to repair pyrimidine dimers.

Inherited loss-of-function mutations can result in a disorder marked by deficiency in NER activity, known as **xeroderma pigmentosum.**

Individuals with xeroderma pigmentosum have extreme photosensitivity, developing severe sunburn from only a few minutes of sunlight exposure (Figure 3). Because t[illegible]ndividuals who lack NER have a significantly increased risk of skin cancer.

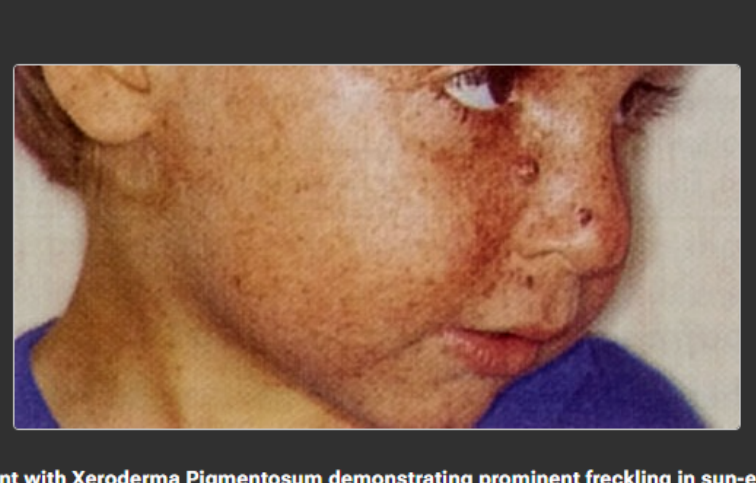


Figure 3. Patient with Xeroderma Pigmentosum demonstrating prominent freckling in sun-exposed areas.

CLINICAL CORRELATION

Squamous cell carcinoma, basal cell carcinoma, and melanoma are all skin cancers associated with chronic sun exposure.

Lynch Syndrome: A Defect in Mismatch Repair

Mismatch repair is designed to correct errors in DNA replication. DNA polymerases aren't perfect, even with their proofreading capabilities. When incorrect bases are incorporated into newly synthesized strands, they are recognized by the "Mut" class proteins that recruit endonucleases. From there, mismatch repair rem[illegible]ide sequence containing the mismatched nucleotide.

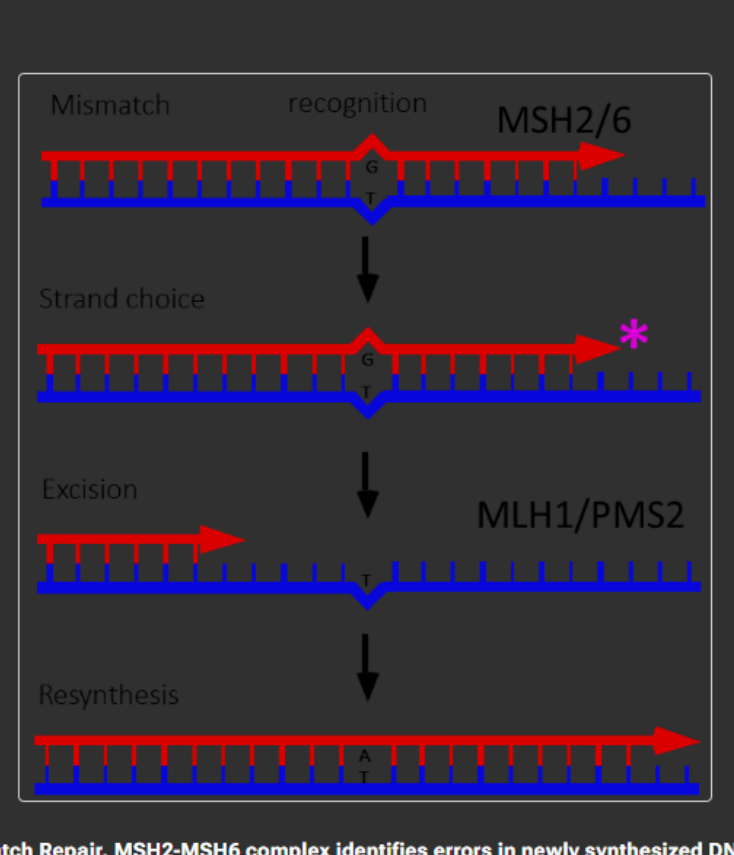


Figure 4. Mismatch Repair. MSH2-MSH6 complex identifies errors in newly synthesized DNA. MLH1-PMS2 excises the incorrect base. The excised strand is filled in by DNA polymerase.

Defects in mismatch repair lead to the accumulation of errors throughout the genome. Some of these errors may create driver mutations and, with time, cause cancer. One of the hallmarks of mismatch repair defects is **microsatellite instability**. Microsatellites are tandem repeats of 1-6 nucleotides found throughout the genome. In normal people, the length of microsatellites remai[illegible]fective, these satellites are unstable and increase or decrease in length.

Defective mismatch repair is a key etiology of **Lynch syndrome**, formerly called hereditary nonpolyposis colorectal cancer (HNPCC). Individuals with Lynch syndrome have significantly increased risk of cancer within the gastrointestinal tract, predominantly in the cecum and proximal colon (Figure 5).

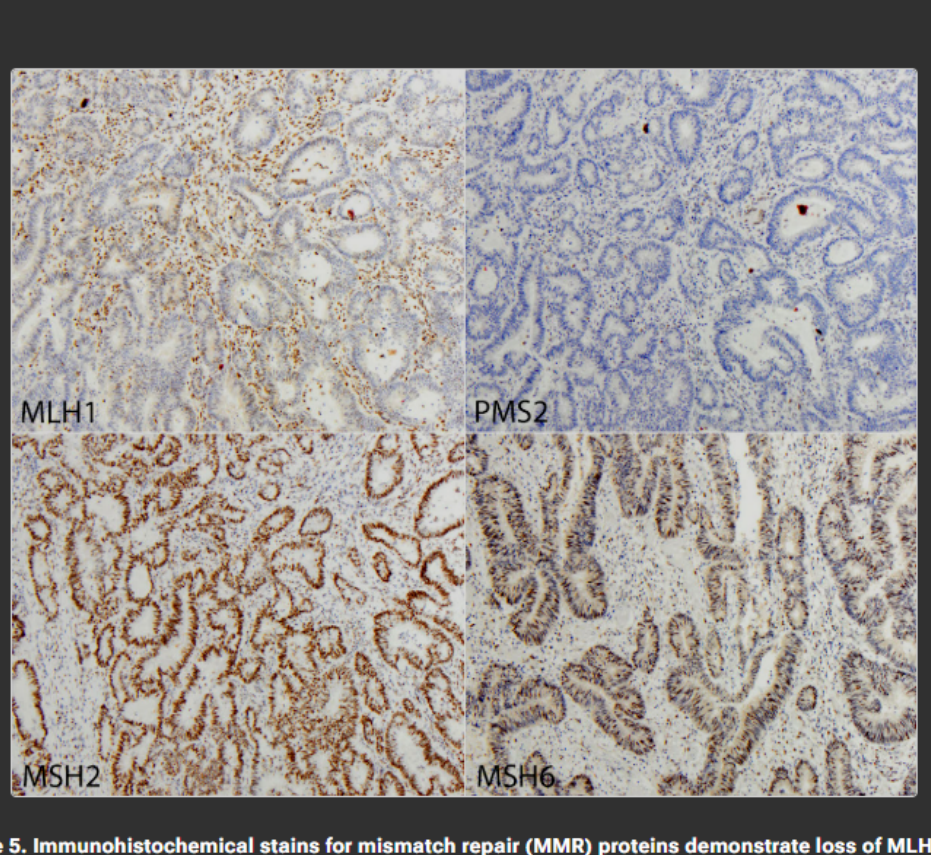


Figure 5. Immunohistochemical stains for mismatch repair (MMR) proteins demonstrate loss of MLH1 and PMS2 staining (lack of brown staining in malignant glands), supporting an MMR-deficient tumor. The bottom two pictures (MSH2 and MSH6) demonstrate normal brown staining of glands, showing intact protein expression.

CLINICAL REASONING

It is typical to observe simultaneous loss of MLH1 and PMS2 protein expression, because these two proteins form a functional heterodimer. This means that a defect in MLH1 usually leads to the destabilization and subsequent loss of PMS2. Therefore, seeing both absent indicates a primary problem with the *MLH1* gene.

Lynch syndrome is the most common inherited colo-rectal cancer susceptibility syndrome and accounts for ~3% of newly diagnosed cases of colo-rectal cancer and 2-3% of endometrial cancer. It is an autosomal dominant disorder caused by a germline mutation in one of several DNA mismatch repair genes: *MLH1*, *MSH2*, *MSH6*, and *PMS2*.

CLINICAL CORRELATION

Clinical signs of possible Lynch syndrome: at least 3 relatives with any Lynch syndrome-associated cancer (colorectal cancer, endometrial cancer, etc.) (at least one should be a first degree relative of the other two), at least two successive generations affected, at least one diagnosed before age 50

CLINICAL CORRELATION

10-15% of sporadic colon cancer cases have microsatellite instability.

What genes are commonly mutated in Lynch syndrome?

Double-Strand Break Repair

DNA helices can sustain damage that actually severs both strands. These lesions are potentially fatal for a cell; if not corrected, the cell will almost certainly undergo apoptosis. Cells have two main mechanisms that attempt to reconnect severed strands (Figure 6).

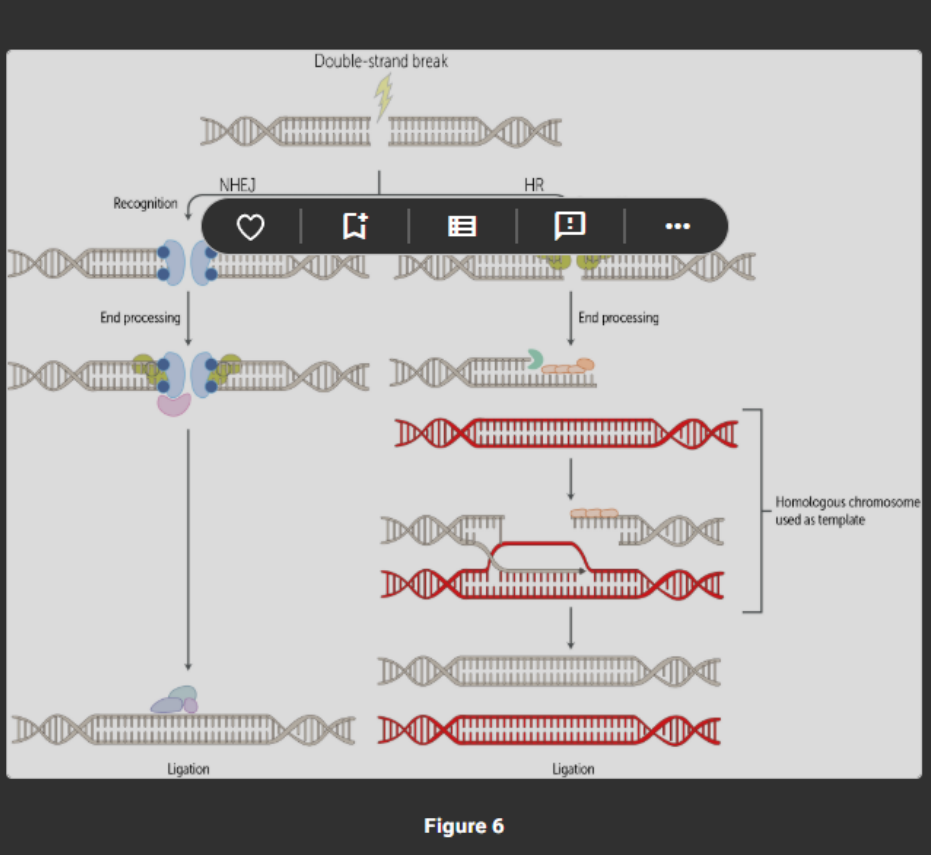


Figure 6

Homologous recombination (HR) uses a homologous chromosome as a template to process the severed strands. When cellular machinery is able to detect two homologous double-strand DNA strands, it will attempt to reconnect the complementary strands without the loss of any nucleotide sequences.

Individuals with ***BRCA1*** (breast cancer gene 1) mutations experience defective HR and as a result are at elevated risk of developing breast and ovarian cancers. Mutations in ***BRCA2*** (breast cancer gene 2) are associated with breast cancer in men and women as well as cancers of the ovary, prostate, pancreas, bile ducts, stomach, melanocytes, and B lymphocytes.

To give you an idea of the effect of *BRCA1/2* mutations on the risk for developing cancer, h[illegible]

- More than 60% of women who inherit a harmful change in *BRCA1* or *BRCA2* will develop breast cancer during their lifetime, in contrast to 13% of women in the general population
- Among women who have been diagnosed with breast cancer, those with *BRCA1* or *BRCA2* mutations have an increased risk of developing cancer in the opposite (contralateral) breast in the future
- About 0.2-1.2% of men with mutated *BRCA1* and 1.8-7.1% of men with mutated *BRCA2* develop breast cancer by age 70, in contrast to 0.1% of men in the general population
- About 39-58% of women with mutated *BRCA1* and 13-29% of women with mutated *BRCA2* develop ovarian cancer during their lifetime, in contrast to 1.1% of women in the general population

CLINICAL CORRELATION

Women with *BRCA1/2* may receive prophylactic mastectomy (removal of the breast) or salpingo-oophorectomy (removal of the Fallopian tubes and ovaries) to lower the risk of cancer.

CLINICAL CORRELATION

Fanconi anemia is an autosomal recessive disorder that may be caused by mutations in more than a dozen different genes, each encoding a protein that participates in a pathway that repairs DNA cross-links through homologous recombination. It is c[illegible]ities (short stature, skeletal abnormalitie[illegible]agents that cross-link DNA, and increased risk of bone marrow failure (aplasia) and leukemia.

What is the role of homologous recombination (HR) in DNA repair?

Nonhomologous end joining (NHEJ) is a mechanism that uses a special ligase complex to directly fuse the ends of two DNA fragments. When HR isn't possible, due to the degree of damage sustained by the DNA or the current status of the cell cycle, the higher priority for the cell is to use NHEJ to reconnect the severed strands, even at the risk of jeopardizing some genetic integrity. Why is HR preferred over NHEJ? Because NHEJ connects any two severed strands it can find (ie, the strands don't have to be homologous), it is error-prone and can result in mutations. If multiple double-strand breaks occur in separate chromosomes, NHEJ can result in translocations; this contributes to the pathophysiolo[illegible]

CLINICAL CORRELATION

The inaccuracy of NHEJ actually has a physiological utility within our immune systems to allow B cells to undergo V(D)J recombination, the process by which novel antibodies and receptors are generated to detect foreign antigens. For this reason, NHEJ defects can result in severe combined immunodeficiency (SCID).

Another clinical consequence of NHEJ defects is ataxia telangiectasia. Individuals with ataxia telangiectasia develop ataxia, chorea, myoclonus, and neuropathy usually with onset before the age of 5 years; they also commonly develop benign blood vessel engorgement within the skin and eyes called telangiectasia (Figure 7). Ataxia telangiectasia also predisposes to cancer, such as leukemia and lymphoma.

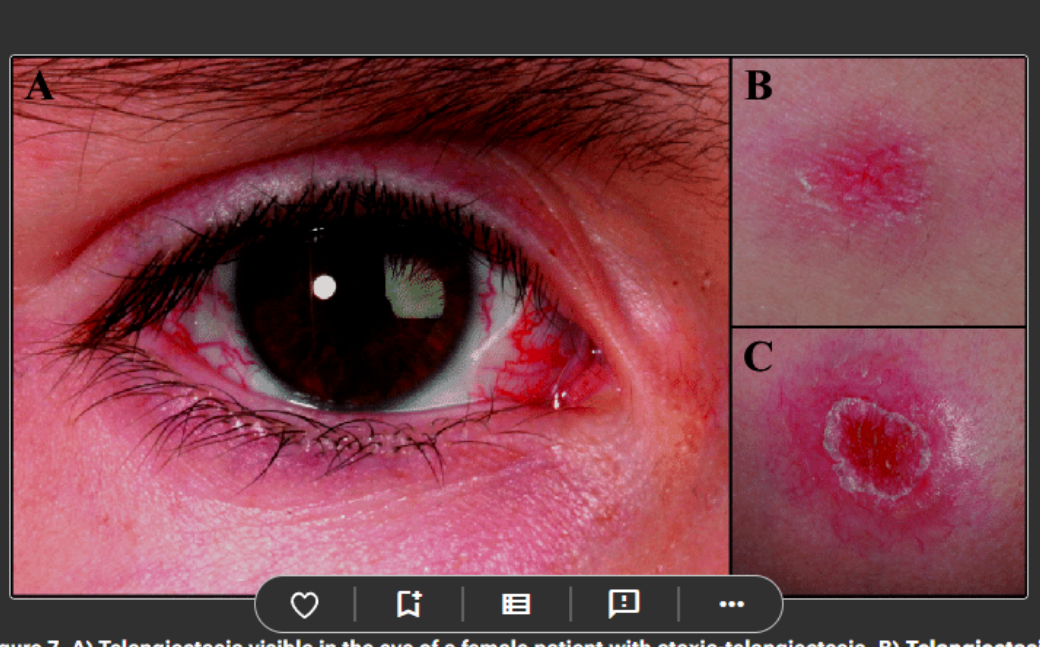


Figure 7. A) Telangiectasia visible in the eye of a female patient with ataxia-telangiectasia. B) Telangiectasia on the skin of the same patient. C) Telangiectasia accompanied by scaling of the skin.

CASE CONNECTION BACK TO INTRODUCTION ↑

You suspect the patient has xeroderma pigmentosum, a disorder marked by deficiency in nucleotide excision repair activity. These patients have extreme photosensitivity, developing severe sunburn after a few minutes of sunlight exposure. Freckling most prominently occurs in sun-exposed areas like the face and neck, while sun-covered areas like the shoulders are relatively protected. These patients are at a significantly increased risk of skin cancer, such as squamous cell carcinoma.

Summary

- DNA lesions that are not repaired become mutations.
- Genomic instability refers to increased propensity of genomic mutations.
- Cells have numerous checkpoints within the cell cycle to assess for DNA lesions and attempt repair; if the degree of genetic damage is too great, apoptosis will be initiated.
- Nucleotide excision repair (NER) removes and replaces an oligonucleotide sequence containing a lesion; defective NER can cause xeroderma pigmentosum.
- Mismatch repair catches errors in DNA replication and occurs primarily in the S phase; defective mismatch repair can cause Lynch syndrome.
- The homologous recombination (HR) mechanism uses a homologous chromosome to a[illegible]aks; the *BRCA1* mutation causes defective HR and predisposes individuals to breast and ovarian cancers.
- Nonhomologous end joining (NHEJ) fuses DNA fragments without requiring homology but is error-prone and can even result in chromosomal translocations; defective NHEJ can lead to severe combined immunodeficiency and ataxia telangiectasia.

Review Questions

1. A patient is found to have inherited a defective protein involved in homologous recombination repair of double-strand breaks. What pathology is the patient at increased risk for developing later in life?

- A. ○ Ataxia telangiectasia
- B. ● Breast or ovarian cancer
- C. ○ Lynch syndrome
- D. ○ Xeroderma pigmentosum

Hide Explanation

The correct answer is breast or ovarian cancer (B). The patient has inherited a *BRCA1* mutation, an etiology of defective homologous recombination repair, predisposing her to breast and ovarian cancer. Ataxia telangiectasia (A) is a condition associated[illegible]. Lynch syndrome (C) is associated with defective mismatch repair. Xeroderma pigmentosum (D) is associated with defective nucleotide excision repair.

2. A patient presents with severe combined immunodeficiency and ataxia telangiectasia. Which DNA repair mechanism is likely defective?

- A. ○ Homologous recombination
- B. ○ Mismatch repair
- C. ○ Nucleotide excision repair
- D. ● Nonhomologous end joining

Hide Explanation

The correct answer is nonhomologous end joining (D). This mechanism is error-prone and can even result in chromosomal translocations. Defective NHEJ can lead to severe combined immunodeficiency and ataxia telangiectasia. Homologous recombination (A) is associated with breast and ovarian cancers due to BRCA1 mutation as well as Fanconi anemia. Mismatch repair (B) is associated with Lynch syndrome. Nucleotide excision repair (C) is associated with xeroderma pigmentosum.

3. Which of the following is a clinical sign of Lynch syndrome?

- A. ● At least three relatives with colorectal and endometrial cancers.
- B. ○ At least two successive generations affected by breast cancer.
- C. ○ A diagnosis of colorectal cancer after age 70.
- D. ○ A family history of only pancreatic cancer.

Hide Explanation

The correct answer is at least three relatives with colorectal and endometrial cancers (A), which are the prototypical cancers associated with Lynch syndrome (there are other associated cancers but at a much lower risk). At least two successive generations should be affected by colorectal or endometrial cancers, not breast cancer (B). A diagnosis of colorectal cancer after age 70 (C) is less likely to be inherited; colorectal cancers diagnosed before age 50 are more suspicious for Lynch syndrome. A family history of only pancreatic cancer (D) is not immediately suspicious of Lynch syndrome, which is more likely associated with colorectal cancer and endometrial cancer.

How would you rate this Brick?

References

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Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01 ☑ ~~Describe the relationship between DNA damage and genomic instability.~~

02 ☑ ~~Recognize human developmental disorders and cancer predisposition syndromes associated with defects in nucleotide excision repair, mismatch repair, homologous recombination, and nonhomologous end joining.~~

PathologyOutlines.com website. https://www.pathologyoutlines.com/topic/colontumormolecular microsatellite.html. Accessed January 2nd, 2025.
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Objective Confidence: 2/2